

t-test

with pooled SD

General
Linear
Model

via

Ordinary Least
Square

+

Wald's
approach to
inference

```
update(pairs(
  emmeans(
    m_ols <- lm(value ~ group, data = d), specs = ~group)), infer = c(TRUE, TRUE))
```

contrast	estimate	SE	df	lower.CL	upper.CL	t.ratio	p.value
group1 - group2	-2.99	1.5	98	-5.97	-0.0142	-1.994	0.0489

```
tidy(t.test(value ~ group, data=d, var.equal=TRUE), conf.int=TRUE) %>% select(...)
```

estimate	df	lower.CL	upper.CL	t.ratio	p.value
-2.99	98	-5.97	-0.0142	-1.99	0.0489

Welch t-test

with Satterthwaite degrees of freedom

General
Linear
Model

via

Generalized
Least Square

+

Satterthwaite
degrees of
freedom

Wald's
approach to
inference

```
update(pairs(
  emmeans(
    m_gls <- gls(value ~ group, weights = varIdent(form = ~ 1 | group), data = d),
    specs = ~group, mode = "satterthwaite")), infer = c(TRUE, TRUE))
```

contrast	estimate	SE	df	lower.CL	upper.CL	t.ratio	p.value
group1 - group2	-2.99	1.5	49.8	-6.01	0.0224	-1.994	0.0517

```
tidy(t.test(value ~ group, data=d, var.equal=FALSE), conf.int=TRUE) %>% select(...)
```

estimate	df	lower.CL	upper.CL	t.ratio	p.value
-2.99	49.8	-6.01	0.0224	-1.99	0.0517